

Formal Verification: Warp Exactly-Once Message Delivery

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Abstract

We prove exactly-once delivery for Warp cross-chain messages. Messages carry BLS aggregate signatures and are processed with nonce-based replay protection. Valid messages succeed on first attempt, replays are rejected, and unauthenticated messages are dropped.

1 Introduction

Warp is the cross-chain messaging protocol for Lux. Messages are authenticated by BLS aggregate signatures from source chain validators. The nonce-based replay protection ensures each message is processed exactly once.

2 Definitions

Definition 1 (WarpMessage). *A Warp message: source chain, destination chain, nonce, payload hash, signature validity.*

Definition 2 (DestState). *Destination state: set of processed nonces and monotonic counter.*

3 Theorems and Axioms

Theorem 3 (no_replay). *A processed nonce is never accepted again.*

Proof. If $\text{nonce} \in \text{processed set}$, `processMessage` returns `none`. □

Theorem 4 (authenticity). *Invalid signatures are rejected: $\text{sigValid} = \text{false} \Rightarrow \text{none}$.*

Theorem 5 (valid_message_succeeds). *A valid message with fresh nonce succeeds: returns `some s'`.*

Theorem 6 (nonces_monotone). *Processed nonce set grows monotonically: $s.\text{nonces} \subseteq s'.\text{nonces}$.*

4 Lean Source

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Warp Cross-Chain Exactly-Once Delivery

Models the Warp messaging protocol used for cross-chain communication.

Warp messages carry BLS aggregate signatures from source chain validators. The destination chain verifies the aggregate signature against the known validator set before processing.

Maps to:

- *src/precompiles/warp/*: Warp precompile at 0x0200...0005
- *treasury/Vault.sol*: usedNonces mapping for replay protection
- *treasury/Collect.sol*: version monotonicity for settings

Properties:

- *Authenticity*: only messages signed by source validators are accepted
- *No replay*: each message processed at most once (nonce tracking)
- *No loss*: valid messages eventually delivered (under liveness assumption)
- *Combined*: exactly-once delivery

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```
import Mathlib.Data.Nat.Defs
import Mathlib.Data.Finset.Basic
import Mathlib.Tactic

namespace Warp.Delivery

/-- Chain identifier -/
abbrev ChainId := Nat

/-- Warp message -/
structure WarpMessage where
  sourceChain : ChainId
  destChain   : ChainId
  nonce       : Nat
  payload     : Nat -- hash of payload
  sigValid    : Bool -- BLS aggregate signature valid

/-- Destination chain state for Warp processing -/
structure DestState where
  processedNonces : Finset Nat -- nonces already processed
  lastProcessed   : Nat       -- monotonic counter

/-- Process a Warp message -/
def processMessage (s : DestState) (msg : WarpMessage) : Option DestState :=
  if !msg.sigValid then none -- reject invalid signature
  else if msg.nonce ∈ s.processedNonces then none -- reject replay
  else some {
    processedNonces := s.processedNonces ∪ {msg.nonce}
    lastProcessed   := max s.lastProcessed msg.nonce
  }

/-- No replay: a processed nonce is never accepted again -/
theorem no_replay (s : DestState) (msg : WarpMessage)
  (hmem : msg.nonce ∉ s.processedNonces) :
  processMessage s msg = none := by
  simp [processMessage]
  intro _
  exact hmem
```

```

/-- Authenticity: only valid signatures are processed -/
theorem authenticity (s : DestState) (msg : WarpMessage)
  (hinv : msg.sigValid = false) :
  processMessage s msg = none := by
  simp [processMessage, hinv]

/-- Valid messages succeed on first attempt -/
theorem valid_message_succeeds (s : DestState) (msg : WarpMessage)
  (hsig : msg.sigValid = true)
  (hfresh : msg.nonce < s.processedNonces) :
  s', processMessage s msg = some s' := by
  simp [processMessage, hsig, hfresh]

/-- Processed nonces grow monotonically -/
theorem nonces_monotone (s : DestState) (msg : WarpMessage)
  (h : processMessage s msg = some s') :
  s.processedNonces < s'.processedNonces := by
  simp [processMessage] at h
  split at h <;> simp_all
  split at h <;> simp_all
  exact Finset.subset_union_left

end Warp.Delivery

```

5 References

Source code: <https://github.com/luxfi/proofs/blob/main/lean/Warp/Delivery.lean>

White papers: <https://papers.lux.network>

Related white papers:

- [lux-warp-messaging.tex](#)
- [lux-interchain-transfers.tex](#)